

Course: Fundamentals of Generative AI and Large Language Models: Theory and Practice

Week 5: Generative Adversarial Networks (GANs): Foundations

Week 5 Outline

- Limitations of VAEs and Motivation for GANs
- Fundamental Idea of Adversarial Learning
- GAN Architecture
 - Generator
 - Discriminator
- GAN Objective and Loss Function
 - GAN Training Procedure
 - GAN Convergence and Equilibrium
- Introduction to DCGAN

Hands-on: Implementation of Basic GAN
Implementation of DCGAN

Today's Session Overview

- Limitations of VAEs and Motivation for GANs
- Fundamental Idea of Adversarial Learning

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Limitations of VAEs

1. Gaussian Assumption:

In a standard VAE, we usually make two important assumptions:
The approximate posterior is usually assumed to be Gaussian:

$$q_{\phi}(z | x) = \mathcal{N}(\mu_{\phi}(x), \sigma_{\phi}^2(x))$$

The prior is generally fixed as:

$$p(z) = \mathcal{N}(0, I)$$

These assumptions make training easier, but real-world data may have a more complicated structure.

2. Blurry Generated Images:

A VAE uses pixel-wise reconstruction loss, such as Mean Squared Error, to compare the generated image with the original image.

$$x^{(1)} \xrightarrow{\text{Encoder}} z^{(1)}$$

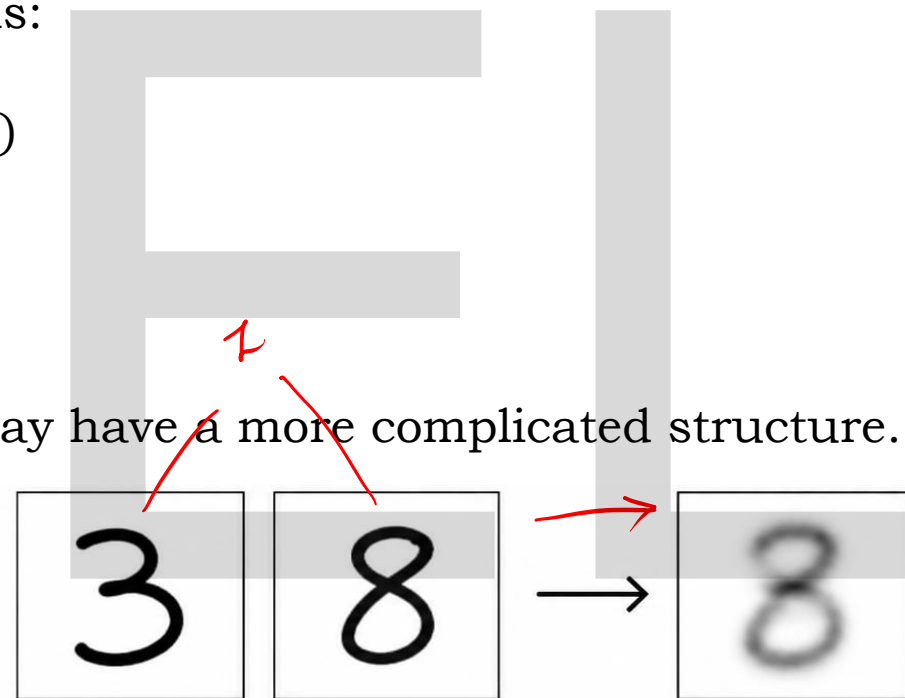
$$x^{(2)} \xrightarrow{\text{Encoder}} z^{(2)}$$

Suppose:

$$z^{(1)} \approx z^{(2)} \approx z$$

The decoder receives
nearly the same latent
representation z

When the model is uncertain between different possible outputs, it may combine or average them instead of generating one clear output. This averaging removes sharp edges and fine details, causing the generated image to appear blurry.



Digit can be "3" or "8"

Limitations of VAEs

3. Trade-off Between Reconstruction and KL Divergence

The VAE loss has two parts:

$$\mathcal{L}_{VAE} = \mathcal{L}_{reconstruction} + D_{KL}(q_{\phi}(z | x) \parallel p(z))$$

These two terms try to achieve different goals.

Reconstruction loss

This term wants the decoder to reconstruct the input accurately:

$$x \approx \hat{x}$$

Therefore, the encoder tries to store more information about the input in the latent vector z .

KL - divergence loss

This term wants the latent distribution to remain close to the standard normal prior:

$$q_{\phi}(z | x) \approx \mathcal{N}(0, I) \checkmark$$

Therefore, it prevents the encoder from storing too much input-specific information. $\checkmark$

Improving one may reduce the performance of the other.

Limitations of VAEs

4. *posterior collapse*

The VAE loss contains:

$$\mathcal{L}_{VAE} = \mathcal{L}_{reconstruction} + D_{KL} \left(q_{\phi}(z | x) \parallel p(z) \right)$$

The KL term encourages:

$$q_{\phi}(z | x) \approx p(z)$$

If this pressure becomes too strong, the easiest way to reduce the KL term is for the encoder to output:

$$\mu(x) \approx 0$$

and

$$\sigma(x) \approx 1$$

for every input.

Then:

$$q_{\phi}(z | x) \approx \mathcal{N}(0, I)$$

and therefore:

$$D_{KL} \approx 0$$

Although the KL loss becomes very small, the latent representation becomes uninformative.

In a **normal VAE**:

$$x \rightarrow q_{\phi}(z | x) \rightarrow z \rightarrow p_{\theta}(x | z) \rightarrow \hat{x}$$

The decoder receives z , and z should contain information about the input x . The decoder uses z to reconstruct

During posterior collapse:

$$q_{\phi}(z | x) \approx p(z)$$

for every input.

This means different inputs produce almost the same latent distribution:

$$x_1 \rightarrow z_1 \approx \mathcal{N}(0, I)$$

$$x_2 \rightarrow z_2 \approx \mathcal{N}(0, I)$$

Therefore, z no longer tells the decoder which specific input was given.

If z does not contain information about x , how can the decoder reconstruct x ?

The answer depends on the type of decoder.

Text: Autoregressive decoder

Images: CNN-based non-autoregressive decoder

Limitations of VAEs

5. *Explicit Likelihood Assumption*

In a VAE, the decoder outputs the parameters of a probability distribution $p_{\theta}(x | z)$. ✓

- For continuous data, the decoder usually outputs the mean ($\mu_{\theta}(z)$) of a Gaussian distribution.
- For binary data, the decoder outputs the probability of each binary value ($\pi_{\theta}(z)$) of a Bernoulli probabilities.

This predefined choice is called the *likelihood assumption*.

This likelihood choice determines the reconstruction loss:



If the selected likelihood is too simple or does not match the true characteristics of the data, then the decoder cannot represent the output uncertainty accurately. As a result, the reconstructed or generated samples may appear less realistic or blurry.

Limitations of VAEs

6. Generated samples may lack fine details

VAEs usually capture the overall structure of the data well, but they may fail to reproduce:

- sharp edges,
- fine textures,
- small facial details,
- realistic high-frequency information.

Therefore, the samples may look smooth but less realistic.

The main motivation for GANs is:

To generate highly realistic data without explicitly defining a probability distribution or reconstruction loss?

Fundamental Idea of Adversarial Learning

Competing with each other

Two neural networks are needed

∴, In a GAN, there are two networks

These two neural networks learn by competing with each other.

G here is Generator

GAN is a neural network used to generate the data

Generator G

The generator generates fake samples from random noise:

$$z \rightarrow G(z) = \hat{x} \quad \checkmark$$

Its goal is:

Generate samples that look real. ✓

Discriminator D

The discriminator receives both real and generated samples and predicts whether each sample is real or fake:

$$D(x) \rightarrow \underline{\text{real or fake}}$$

Its goal is:

Correctly distinguish real samples from generated samples.

Fundamental Idea of Adversarial Learning

How the competition works

The **discriminator** learns to distinguish real samples from fake samples generated by the generator.

At the same time, the **generator** learns to generate realistic samples so that the discriminator cannot easily identify them as generated fake images.

So the learning process is:

Generator *generates a fake sample*



Discriminator *classifies it as real or fake*



The Generator uses the discriminator's prediction to improve



The process is repeated

Through this competition, the generator gradually learns to generate more realistic samples, while the discriminator becomes better at identifying fake samples.

Next Session

GAN Architecture

- Generator
- Discriminator

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Thank you